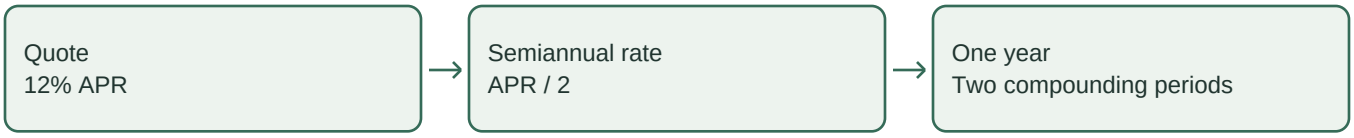
Rate quotes, loan ledger & purchasing power



A. Rate quotes describe a convention; convert to an effective rate for the cash-flow period.

A. Savings quote

Deposit today	$1,000.00
Quoted APR	12%
Compounding	Semiannual (twice per year)
Withdrawals, tax, fees	None

Compute a half-year balance first, then a full-year balance. Reinvest all interest. APR and EAR coincide only when compounding is annual (or the rate is zero).

B. Loan statement

Opening principal before month 1	$2,000.00
Quoted APR	12%, compounded monthly
Scheduled month-end payment	$220.00
Extra payment in month 1	None
Extra payment in month 2	$100.00 after the scheduled payment
Fees or penalties	None

Monthly rate = APR / 12. Interest accrues on the opening balance. Each scheduled payment first covers that month's interest; its remainder reduces principal. The extra month-2 payment reduces principal immediately.

C. Borrower decision note

Outstanding balance	PV of remaining contractual payments at the loan rate
Refinance comparison	Include fees, remaining term, and all changed payments
Teaser rate	An initial rate may reset later

A lower monthly payment alone does not establish a lower financing cost; a longer repayment term may explain the reduction.

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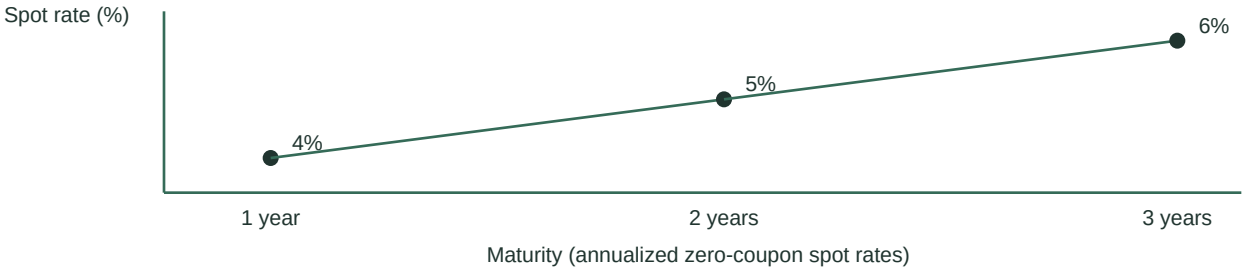

D. Spot-rate valuation

Payment in one year	Discount by 1.04
Payment in two years	Discount by 1.05 squared
Payment in three years	Discount by 1.06 cubed

These are annualized zero-coupon spot rates, not sequential one-year rates. A flat-rate annuity shortcut does not apply directly to this nonflat curve. This is an illustration, not a current market quote.

E. After-tax purchasing-power case

Deposit today	$1,000.00
Nominal effective annual interest rate	10%
Tax rate on interest only	20%, paid at year end
Inflation over the year	8%
Principal tax, fees, other flows	None

Compute gross interest, tax on that interest, after-tax ending dollars, then divide by the inflation factor to express the result in today's purchasing power. Carry full precision until the end.

F. Discount-rate matching

Cash-flow attributes	Currency, date, risk, nominal/real, before/after tax
Opportunity cost	Expected return on a comparable-risk, comparable-term alternative
Promised risky yield	May exceed expected return because of default losses
Interest tax deduction	Use only when explicitly assumed usable

A risky portfolio's expected return is not a suitable discount rate for a guaranteed liability merely because the portfolio is intended to fund it. Retained cash also has an opportunity cost.